Seed label _____ Date _____

Source D. An auditor for D may see this sheet, never sheet C or E.

VOIDFORGE METHOD · SHEET D · DICE 12-word seed

1. **D1 · ≈7 min.** Throw two dice from the same set in a cup (or one die twice). Read the **left** die first. Write both numbers in the next pair box. If they are 1-1, 2-2, 3-3 or 4-4, cross the box out ("again"). Otherwise find the cell in the mirror table (row = first number, column = second) and write its 5 bits into the next shaded run of grid D (runs are numbered). Stop when cell 128 is full (the last run uses only 3 bits).
2. **D2 · ≈1 min.** Bias box: tally the faces of the **first 26 pair boxes** (52 throws, crossed-out boxes included) and count repeats (a throw equal to the throw just before it, across box edges too).
3. **D3 · ≈1.5 min.** Marks. Row mark R: write 1 at the end of a row if the row has an odd number of 1s, else 0. Column mark M: the same for each column, written in row M.
4. **D4 · optional.** Seed XOR part: words 1-11 of D are read like sheet E; a tool adds D's own last word. Never type D into the device that will hold E unless you accept that it learns D.

1st \ 2nd	1	2	3	4	5	6
1	*again*	0000**1**	0001**1**	0010**1**	0011**1**	0100**1**
2	0000**0**	*again*	0101**1**	0110**1**	0111**1**	1000**1**
3	0001**0**	0101**0**	*again*	1001**1**	1010**1**	1011**1**
4	0010**0**	0110**0**	1001**0**	*again*	1100**1**	1101**1**
5	0011**0**	0111**0**	1010**0**	1100**0**	1111**0**	1110**1**
6	0100**0**	1000**0**	1011**0**	1101**0**	1110**0**	1111**1**

Mirror rule: cells (a,b) and (b,a) share the first 4 bits; the bold 5th bit is 1 when the second throw is higher. 5-5 ends in 0, 6-6 in 1. For one die thrown twice this bit is exactly fair even if the die is biased (classes 5-5/6-6 excepted). The other 4 bits are fair only for a fair die.

Bias box (D2) — detects gross defects only

face	tally	count
1		
2		
3		
4		
5		
6		
repeats		

FAIL if any count ≥ 18 or ≤ 1, or repeats ≤ 2.
Fair dice fail about 2% of the time. On FAIL: new dice, new sheet D. Never edit bits.

Result: PASS ☐ FAIL ☐

PAIR BOXES (FIRST THROW | SECOND THROW)

1	2	3	4	5	6	7	8	9	10	
11	12	13	14	15	16	17	18	19	20	
21	22	23	24	25	26	27	28	29	30	
31	32	33	34	35	36	37	38	39	40	

GRID D

	1	2	3	4	5	6	7	8	9	10	11	R
1	1					2					3	
2					4					5		
3				6					7			
4			8					9				
5		10					11					
6	12					13					14	
7					15					16		
8				17					18			
9			19					20				
10		21					22					
11	23					24					25	
12					26							
M												

Shaded and white runs alternate: each run is one accepted pair (5 bits). Small numbers = run number. Hatched cells are not used.

Auditor D re-reads every pair box in the mirror table and compares each run, re-tallies the bias box and recomputes R and M. Then signs:

M(D) and R(D) checked by _____

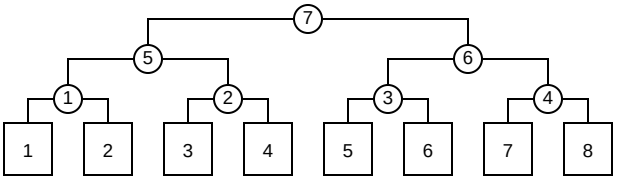
Seed label _____ Date _____

Source C. An auditor for C may see this sheet, never sheet D or E.

VOIDFORGE METHOD · SHEET C · CARDS 12-word seed

1. **C1 · ≈1.4 min per deck.** Shuffle one full 52-card deck: spread it face down and wash (stir) it for 30 seconds, gather, then **12 riffle shuffles** and one cut. 3 decks in all (you may reuse the same deck).
2. **C2 · ≈2.5 min per deck.** Deal 8 cards face up in a row, left to right. Play the 7 matches in the numbered order below. In every match write **1 if the RIGHT card is higher**, 0 if the left card is higher, into the next run of grid C (labels: deck.octet). After a match, slide the lower card down; the higher card plays on. After 6 octets, deal 4 cards (quad, label *q*: matches 1v2, 3v4, then the two winners).
3. **C3 · while dealing.** Count **neighbours**: two cards side by side in a group, same suit, ranks next to each other (7♥ 8♥, 8♥ 7♥, A♣ 2♣, Q♠ K♠). After each deck count the 1s it gave.
4. **C4 · ≈1.5 min.** Marks R and M, as on sheet D. *Audit mode* (+≈4.5 min): also write every card (e.g. 7H, TS, KC) in the record below so auditor C can redo the matches.

Card order: A 2 3 4 5 6 7 8 9 10 J Q K (A low, K high). Same rank: ♣ < ♦ < ♥ < ♠ (alphabetical: clubs, diamonds, hearts, spades). There are no ties.



Matches ①–④: 1v2, 3v4, 5v6, 7v8. ⑤: winner of 1v2 against winner of 3v4. ⑥: winners of 5v6 and 7v8. ⑦: the final, left half against right half. Write the bits in this order.

Bias box (C3) — detects gross defects only

deck	1s	neighbours	pass?
1			
2			
3	—		

FAIL a deck if its 1s are ≤ 11 or ≥ 34 (full 45-bit decks only), or neighbours ≥ 6. A good shuffle fails about 0.6% of decks. On FAIL: reshuffle and redo that deck's runs.

GRID C

	1	2	3	4	5	6	7	8	9	10	11	R
1	1.1							1.2				
2				1.3							1.4	
3							1.5					
4			1.6							1q		
5		2.1							2.2			
6					2.3							
7	2.4							2.5				
8				2.6							2q	
9			3.1							3.2		
10						3.3						
11		3.4							3.5			
12					3q							
M												

Runs alternate shaded and white: an octet gives 7 bits, a quad 3. Label 2.3 = deck 2, octet 3; 1q = deck 1 quad.

Deck plan: deck 1: 6 octets + 1 quad; deck 2: 6 octets + 1 quad; deck 3: 5 octets + 1 quad.

Auditor C redoes the matches from the record (audit mode), recounts the bias box and recomputes R and M. Then signs:

M(C) and R(C) checked by _____

CARD RECORD (AUDIT MODE)

deck 1 oct 1								
deck 1 oct 3								
deck 1 oct 5								
deck 1 quad								
deck 2 oct 2								
deck 2 oct 4								
deck 2 oct 6								
deck 3 oct 1								
deck 3 oct 3								
deck 3 oct 5								

deck 1 oct 2								
deck 1 oct 4								
deck 1 oct 6								
deck 2 oct 1								
deck 2 oct 3								
deck 2 oct 5								
deck 2 quad								
deck 3 oct 2								
deck 3 oct 4								
deck 3 quad								

Seed label _____ Date _____

SECRET: grid E is your seed. Only the owner sees this sheet. Destroy it after the backup.

VOIDFORGE METHOD · SHEET E · FORGE 12-word seed

1. **E1 · ≈2.5 min.** XOR. Put sheets D and C side by side. For each cell: same bits → **0**, different → **1**. Write it in grid E.
2. **E2 · ≈2 min.** Forge seal. Write R and M for grid E. Copy the marks of D and C into the seal table, XOR them and compare with E's marks. All must agree; a mismatch points to the row and column of the mistake.
3. **E3 · ≈3 min.** Words. Read each full row as 3-4-4: **block** (bits 1-3), **row** (4-7), **column** (8-11) in the VOIDFORGE binary word table; no arithmetic. (Numbered 1-2048 list instead: add the weights above the 1-cells, plus 1.)
4. **E4 · ≈2 min.** Check the words. Find each word alphabetically in the table, read its bits from the headings and tick ✓ if they equal the grid row.
5. **E5 · ≈4 min.** Last word (box ⑤). It needs SHA-256: two independent tools must agree and the word must lie in the range you found by hand. Then compare fingerprint and first address.
6. **E6 · ≈2 min.** Backup. Copy the 12 words **and** marks M and R (optional: skew mark K) to the backup. Keep marks only next to the words: on their own they reveal 23 bits.

GRID E = D ⊕ C

	1024	512	256	128	64	32	16	8	4	2	1			
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**R**	**word**	✓
1	*a*	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*	*j*	*k*			
2	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*	*j*	*k*	*a*			
3	*c*	*d*	*e*	*f*	*g*	*h*	*i*	*j*	*k*	*a*	*b*			
4	*d*	*e*	*f*	*g*	*h*	*i*	*j*	*k*	*a*	*b*	*c*			
5	*e*	*f*	*g*	*h*	*i*	*j*	*k*	*a*	*b*	*c*	*d*			
6	*f*	*g*	*h*	*i*	*j*	*k*	*a*	*b*	*c*	*d*	*e*			
7	*g*	*h*	*i*	*j*	*k*	*a*	*b*	*c*	*d*	*e*	*f*			
8	*h*	*i*	*j*	*k*	*a*	*b*	*c*	*d*	*e*	*f*	*g*			
9	*i*	*j*	*k*	*a*	*b*	*c*	*d*	*e*	*f*	*g*	*h*			
10	*j*	*k*	*a*	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*			
11	*k*	*a*	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*	*j*			
12	*a*	*b*	*c*	*d*	*e*	*f*	*g*						last word: see box ⑤	
M														

Small italic letters are for the optional skew mark K (count the 1s under each letter; odd → 1). K lets an heir also catch two words written in swapped places.

② Forge seal (E2)

M(D)											
M(C)											
M(D ⊕ C)											
M(E)											

R(D)												
R(C)												
R(D ⊕ C)												
R(E)												

Skew mark K (optional)

	a	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*	*j*	*k*
1s											
K											

⑤ Last word and identity (E5)

Last 7 bits of grid E	
Word 12 must be between	_____ and _____ bits 1-3 = block, 4-7 = row: one of that row's 16 words
Tool A (VOIDFORGE Word draw)	
Tool B (SeedSigner Calc final word)	
A = B, and inside the range?	yes ☐ no ☐ → stop and find the mistake
Fingerprint A / B / wallet	_____ / _____ / _____
First address A / wallet (first 6 + last 6)	_____ / _____